

Digital Signal Processing exam

July 10, 2024

1 Theoretical questions

1. Provide examples of physical representations of a signal.
2. Compute the Fourier transform of the Dirac comb?
3. What do we mean by the Nyquist limit?
4. Compare the anti-aliasing and reconstruction filters.
5. What is impulse response and what is its relevance to LTI systems?
6. What is a decibel?
7. Describe the signal characterized by a Laplace transform with a pole
 - (a) of negative/positive real part;
 - (b) having an imaginary component;
 - (c) lying on the imaginary/real axis;
 - (d) in the origin.
8. Choose two of the mapping techniques for designing IIR filters and describe them. (direct design, bilinear transform, impulse-invariant, pole-zero mapping)?
9. Explain the relationship between wavelet functions, scaling functions and filters.

2 Problems

1. Test the systems below for linearity, time-invariance, causality, memory and stability ($x[n]$ is the input):
 - (a) $y[n] = x[n]u[n]$

(b) $y[n] = \prod_{k=0}^5 a[k]x[n-k]$

2. Define the discrete-time signal $x(n)$ as

$$x(n) = -0.3\delta(n+2) + 2.0\delta(n) + 1.5\delta(n-3) - \delta(n-5)$$

- (a) Sketch $x(n)$.
- (b) Write the Z -transform $X(z)$.
- (c) Define $G(z) = z^{-2}X(z)$. Sketch $g(n)$.

3. The ideal continuous-time band-stop filter has the frequency response

$$H^f(\omega) = \begin{cases} 1 & |\omega| \leq \omega_1 \\ 0 & \omega_1 < |\omega| < \omega_2 \\ 1 & |\omega| \geq \omega_2 \end{cases}$$

- (a) Sketch $H^f(\omega)$.
 - (b) What is the impulse response $h(t)$ of the ideal band-stop filter?
 - (c) Describe how to implement the ideal band-stop filter using only lowpass and highpass filters.
4. A stable linear time invariant system has the transfer function

$$H(z) = \frac{z(z+2)}{(z-1/2)(z+4)}$$

- (a) Find the frequency response $H^f(\omega)$ of this system.
- (b) Calculate the value of the frequency response $H^f(\omega)$ at $\omega = 0.2\pi$.
- (c) Find the output $y(n)$ produced by the input $x(n) = \cos(0.2\pi n)$.

3 Programming tasks

1.
 - (a) Generate a finite support, symmetric random signal.
 - (b) Interpreting it as the Fourier transform (in frequency space) of some other signal, find the original signal.
 - (c) Demonstrate the effect of sampling on this derived signal.
 - (d) Starting from the sampled signal reconstruct the original (continuous) signal and demonstrate the effect of the Nyquist theorem.
2.
 - (a) Read in and plot the signal from the **data.bin** file (it is in a contiguously laid out binary format).
 - (b) Determine the nature/origin of the data and determine its sampling rate. Justify!
 - (c) Represent the signal's spectrogram
 - (d) Represent the signal's continuous wavelet transform.
 - (e) Identify& describe a few elements oscillatory elements in the signal based on the above time-frequency representations.
 - (f) Identify a band of practical relevance and apply a bandpass filter over that band. Demonstrate the effect of the filtering via the power spectral density, the spectrogram. Save the filtered signal in a directly usable format.
3.
 - (a) Design a digital band-pass filter with a lower cutoff frequency of 1 kHz and an upper cutoff frequency of 3 kHz using the bilinear transform method. Specify the desired sampling frequency and provide the filter coefficients. Compare the theoretical frequency response (gain & phase) with the numerical one (use the **signal** Python module.)
 - (b) (+10%) Demonstrate the usage of frequency pre-warping.